



Lesson 8.1 — Statistical Questions and Data

Big idea: A statistical question expects variability in answers and is investigated with data. Knowing what KIND of data you have decides what tools you'll use.

Quick review

Statistical question: one whose answer varies (e.g., “How tall are 10th graders?”). NOT statistical: “How tall is Maya?”

Categorical (qualitative) data: labels or categories (eye color, brand). Summarize with counts/percents and bar charts.

Quantitative data: numbers (height, hours of sleep). Summarize with mean, median, spread; visualize with histograms, dot plots, box plots.

Population vs. sample: the population is everyone you care about; the sample is the subset you actually measure.

Statistic vs. parameter: sample mean (statistic) ESTIMATES population mean (parameter).

Worked example

Worked example. For each, classify as statistical/non-statistical and (if quantitative) the variable type.

- (a) “What is the average GPA of seniors?” → statistical, quantitative.
- (b) “What is Tomás’s favorite color?” → not statistical (single answer).
- (c) “What types of pets do students have?” → statistical, categorical.

Warm-ups

1. Is “How many hours of sleep do 9th graders get?” a statistical question?
2. Is data on eye color categorical or quantitative?
3. Identify population vs. sample: surveying 100 students at one school to estimate the average GPA of all students at that school.

Practice

1. Classify: number of siblings (categorical or quantitative)?
2. Classify: zip code — is it quantitative? Why or why not?
3. Identify the statistic vs. parameter in: “The mean salary of 200 sampled employees was \$52,000; the true mean for all employees is \$54,500.”

Challenge

1. Design a statistical question to investigate study habits of high school students. Specify the population, the variable(s), and how you’d collect data.
2. Why might a survey distributed only on social media give a biased estimate of average teen TV viewing? Describe the bias.



Answer key — Lesson 8.1

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. Yes (variability expected).
2. Categorical.
3. Population: all students at the school. Sample: the 100 surveyed.

Practice

1. Quantitative.
2. It's NUMERICAL but treated as CATEGORICAL — the digits don't carry numerical meaning (you don't average zip codes).
3. Statistic: \$52,000 (sample). Parameter: \$54,500 (population).

Challenge

1. Many possible. Example: "How many hours per week do 11th graders at our school spend studying?"
Population: 11th graders at our school. Variable: hours/week (quantitative). Method: random sample, anonymous survey.
2. Selection bias: teens who use social media (and respond) are not representative of all teens — they likely watch differently from those who avoid social media. Their reports under- or over-estimate the population mean.